Vivekanand Education Society's Institute of Technology

Department of Computer Engineering

Group No. : 18
Date :- 11-8-2025

Project Synopsis Template (2025-26) - Sem VII

Ingro: *Simplifying Groundwater Intelligence from INGRES*

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Abstract

- The proposed solution is an AI-powered chatbot integrated with the INGRES (India's Groundwater Resources Estimation System) portal, designed to simplify access to groundwater data and assessments. Current GIS-based systems require technical expertise and manual navigation, creating barriers for farmers, policymakers, and researchers.
- The chatbot enables natural language queries, removing the complexity of data retrieval and enhancing accessibility for non-technical users.
- It leverages INGRES Layers (business data, hierarchy, and map business data) to fetch both real-time and historical groundwater results, ensuring up-to-date insights through Retrieval-Augmented Generation (RAG).
- The system also auto-generates visual dashboards—such as rainfall trends, recharge vs. extraction graphs, and groundwater stage maps—based on user queries.
- Further, the chatbot supports multilingual and voice-enabled interactions (Hindi, Marathi, Telugu, etc.), bridging the accessibility gap for rural users unfamiliar with English-only interfaces.
- Built using Python, FastAPI, and React, with integration of VAPI for speech and vector databases for contextual retrieval, the solution enhances the usability of the INGRES portal, enabling data-driven decision-making for government officials, planners, NGOs, and citizens.
- By combining conversational AI, real-time APIs, and contextual visualization, the system represents a unique step towards democratizing groundwater intelligence in India.

Introduction

- **Context of Groundwater Data Access:** Groundwater plays a critical role in India's agriculture, drinking water supply, and rural livelihoods, but accessing its assessments is highly technical. The **INGRES portal** hosts detailed datasets and GIS maps on recharge, extraction, and development stages, yet non-technical users like farmers, NGOs, and local officials struggle to interpret this information.
- **Accessibility and Usability Gaps:** Navigating GIS layers, datasets, and technical jargon creates barriers for decision-making at the grassroots level. Farmers may not know whether they can safely dig a borewell, plant certain crops, or if an area is overexploited. Likewise, policymakers often spend considerable time retrieving past and current groundwater reports, slowing down planning and regulation.

- **Project Objective:** The proposed solution is an **AI-powered multilingual chatbot** integrated with INGRES. It allows users to query groundwater conditions in **natural language** and automatically generates **visual dashboards**—such as rainfall trends, recharge vs. extraction ratios, and groundwater stage classifications. The chatbot fetches real-time and historical results through **INGRES Layers**, ensuring up-to-date and reliable insights.
- **Innovation and Technical Edge:** Unlike static portals, this solution uses **Retrieval-Augmented Generation (RAG)** and vector databases for contextual retrieval, combined with **voice-enabled, multilingual support** (Hindi, Marathi, Telugu, etc.). This makes groundwater intelligence available not just to researchers but also to rural communities in their native language.
- **Scope and Feasibility:** Developed using **Python, FastAPI, React, vector databases, and INGRES APIs**, with VAPI for voice integration, the solution is technically feasible within hackathon timelines. It builds directly on available APIs, ensuring sustainability and scalability.
- **Impact and Accessibility:** By democratizing groundwater insights, the project enables **data-driven decision-making** for farmers, NGOs, and government authorities. It directly supports sustainable water use, reduces over-extraction risks, and strengthens participatory water governance at both local and national levels.

Problem Statement

- Groundwater is India's most critical resource for agriculture, drinking water, and rural livelihoods.
- The **INGRES portal** provides detailed assessments of groundwater recharge, extraction, and development, but its GIS-heavy interface and technical datasets make it difficult for farmers, NGOs, and even local officials to access or interpret.
- This leads to poor decision-making—such as digging borewells in overexploited zones or mismanaging water resources—worsening India's water crisis.
- There is a need for a user-friendly, multilingual, and intelligent system that can make groundwater data easily accessible, understandable, and actionable.

Proposed Solution

- We propose an AI-powered multilingual chatbot integrated with the INGRES portal that enables natural language queries about groundwater.
- The chatbot will fetch both real-time and historical assessments through INGRES APIs and present insights in simple language along with auto-generated visual dashboards (rainfall, recharge vs. extraction, groundwater stage maps).

- Using Retrieval-Augmented Generation (RAG) and voice support, the solution ensures up-to-date, accurate, and inclusive access for diverse stakeholders.
- This makes the INGRES portal more intuitive, empowering farmers, policymakers, researchers, and NGOs to make sustainable, data-driven water management decisions.

Methodology

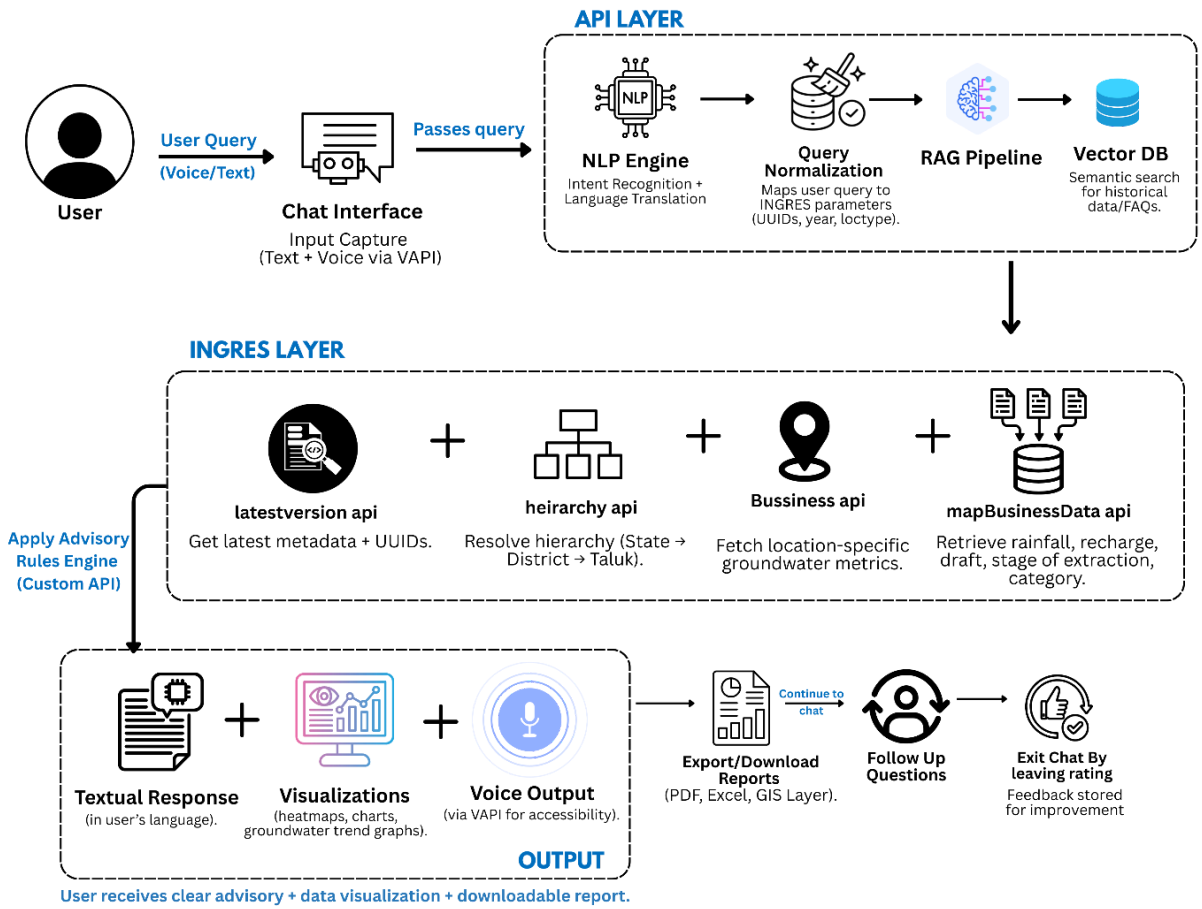

Fig 1. Methodology Diagram

The methodology outlines the systematic approach to develop an **AI-powered multilingual chatbot integrated with the INGRES portal** for simplifying access to groundwater data. It aligns with sustainable water governance goals by providing real-time, historical, and contextual insights through INGRES Layers.

Requirement Analysis:

- Study INGRES Layers (business data, hierarchy, map business data) to understand data formats, coverage, and response structures.
- Identify user requirements for farmers, NGOs, and policymakers (e.g., borewell feasibility, groundwater trends, district-level reports).
- Define chatbot features: natural language querying, voice/multilingual support, auto-generated dashboards, and role-specific access (public vs. authority).

System Design:

- Architect the solution with a **React frontend** for conversational UI and dashboards, and a **Python/FastAPI backend** for API orchestration.
- Design modules: query processor (NLP + RAG), INGRES API connector, visualization engine, multilingual/voice handler, and access control for authorities.
- Use a vector database (e.g., Pinecone, FAISS) to store embeddings of INGRES metadata for contextual retrieval.

Development Phase 1: Query Understanding & Data Retrieval

- Implement natural language understanding using an LLM (e.g., GPT/LLM via LangChain) to parse groundwater-related queries.
- Connect backend to INGRES APIs for fetching real-time and historical groundwater assessments.
- Build a retrieval pipeline (RAG) combining user queries with INGRES API data for accurate responses.

Development Phase 2: Visualization & Insights

- Integrate **Matplotlib/Plotly/D3.js** for generating contextual charts (rainfall trends, recharge vs. extraction, groundwater stage heatmaps).
- Enable dynamic dashboard creation where chatbot auto-generates visual insights alongside text.
- Provide downloadable reports for authority-level users (PDF/Excel).

Development Phase 3: Multilingual & Voice Support

- Add multilingual support using MarianMT/IndicNLP for Hindi, Marathi, Telugu, etc.
- Integrate **VAPI or similar voice API** for speech-to-text (query input) and text-to-speech (response output).
- Localize technical terms (e.g., "stage of development") into user-friendly regional language equivalents.

Development Phase 4: User Interface & Workflow

- Create a conversational chatbot interface (React) with options for text, voice, and visualization output.
- Implement authority dashboards to generate regional and consolidated groundwater reports.
  Ensure accessibility with simple prompts and context-aware suggestions for non-technical users.

Development Phase 5: Analytics & Governance

- Log query trends (e.g., most common queries by farmers vs. officials) for usage insights.
- Provide anonymized usage data to INGRES for future improvements.
- Enable authority users to compare multi-year assessments for planning and policy decisions.

Testing and Validation

- Unit test query classification (farmer queries vs. authority queries) and accuracy of groundwater data retrieval.
- Integration testing of chatbot with INGRES Layers for different datasets.
- User validation with real-world scenarios (e.g., "Can I dig a borewell in Village X?") to ensure accurate and practical responses.

Deployment and Documentation

- Deploy backend APIs on a scalable cloud (AWS/GCP) and host the chatbot frontend for public and authority use.
- Containerize services using Docker for portability and reliability.
- Publish project documentation, API references, and a demo video showcasing farmer and policymaker use cases.

Hardware , Software and tools Requirements

Hardware:

- **Development:** Laptop/PC (8GB RAM, 2.0 GHz CPU, 50GB storage).
- **User:** Smartphone or laptop with internet access, microphone (for voice queries), and modern browser.
- **Server:** Cloud instance (AWS t2.micro / GCP e2-medium, 1–2 vCPUs, 2–4 GB RAM, 20GB storage).

Software:

- **OS:** Windows/Linux/macOS (development); any OS with browser (client).
- **Frontend:** React 18.x for chatbot UI + dashboards.
- **Backend:** Python 3.10+, FastAPI 0.95+ for API orchestration.
- **Database:** PostgreSQL / MongoDB Atlas (for logs, embeddings, query history).
- **Visualization:** Plotly/D3.js/Matplotlib for charts & maps.
- **Multilingual Support:** MarianMT / IndicNLP models for translation.
- **Voice Support:** VAPI / WebRTC + Speech-to-Text (Whisper API).

Tools/Libraries:

- **IDE:** VS Code / PyCharm.
- **Version Control:** Git, GitHub.
- **NLP/AI:** LangChain, HuggingFace Transformers, FAISS/Pinecone (vector DB).
- **APIs:** INGRES APIs (business data, hierarchy, map data).
- **Security:** JWT for session handling, HTTPS/TLS with OpenSSL.
- **Deployment:** Docker, Kubernetes (if scaling), AWS/GCP/Azure

Proposed Evaluation Measures

- **Query Understanding Accuracy:** ≥90% correct classification of farmer vs. authority queries.
- **Data Retrieval Reliability:** 100% successful fetch of real-time & historical results from INGRES APIs.
- **Visualization Quality:** Auto-generated charts/maps validated for correctness against raw INGRES data.
- **Multilingual Support:** ≥95% translation accuracy for Hindi, Marathi, Telugu queries.
- **Voice Query Success Rate:** ≥90% successful speech-to-text conversions in user testing.
- **Authority Reports:** 100% accurate generation of district/regional-level downloadable reports.

Conclusion

- **Effective Solution:** The chatbot bridges the gap between **technical groundwater datasets** and **user-friendly insights**, enabling conversational access to INGRES.
- **Innovation:** Uses **RAG + APIs + contextual dashboards** instead of static text responses, making data more interactive and actionable.
- **Accessibility:** Multilingual and voice-enabled design ensures usability for farmers, NGOs, and rural communities.
- **Feasibility:** Achievable within SIH hackathon timelines using open-source tools (Python, React, LangChain, FastAPI).

- **Impactful Outcome:** Enhances **data-driven water governance**, prevents misuse of groundwater, and empowers sustainable decision-making at all levels.

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